CHAPTER ONE

FUNDAMENTALS OF BIOLOGICAL BEHAVIOUR

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1. INTRODUCTION

Life, in order to persist, requires a continuous throughput of (1) matter and (2) energy:

(1) The most prominent elements passing through the life system are hydrogen (H), oxygen (O), carbon (C), nitrogen (N), phosphorus (P) and sulphur (S), plus a host of minor elements.

(2) Energy is derived principally from three sources and these form a convenient criterion to categorise the organisms implicated:

(a) Solar radiation: photo-synthetic autotrophs

(b) Organic compounds: heterotrophs

(c) Inorganic compounds: chemical autotrophs.

(a) In the biosphere the fundamental source of energy is solar radiation. The photo-synthetic autotrophic cells fix a small fraction of the solar energy by forming complex high energy organic (H, C, O) compounds and producing oxygen. Of the matter requirements the elements H and O are obtained from H_2O; carbon from CO_2; and phosphorus, nitrogen, sulphur and the minor elements from the dissolved salts of these elements. Both H_2O and CO_2 are readily available; sulphur is usually in adequate supply; however the availability of phosphorus and nitrogen usually is limited – phosphorus does not occur readily in soluble form and nitrogen is useful to the organism principally in the NH_4^+ and NO_3^- forms. The restricted availability of these two elements usually constitutes the limiting factor to the mass of autotrophic life a body of water can generate. For this reason, phosphorus and nitrogen are termed eutrophic (life-giving) substances. Phosphorus is limited by the mass available or entering a particular ecosystem, but ammonia can be generated from dissolved molecular nitrogen by certain micro-organisms. In this respect, the control of phosphorus in the body of water takes on a greater significance than the control of nitrogen.

Whereas the eutrophic elements N and P normally are the principal agents of pollution in bodies of water, by stimulating autotrophic growth, the action associated with this growth, of producing oxygen, is utilized in the oxidation pond system of wastewater treatment – the oxygen is utilized by heterotrophic organisms to metabolize the carbonaceous material in the influent to the pond, in this fashion to assist in destroying organic energy, as described in (b) below.

(b) The high energy organic compounds synthesized by the autotrophs form the basic source of energy for the heterotrophic cells to synthesize the more complex molecules that constitute the cell mass, including proteins. Only a fraction of the energy utilized is incorporated in the cell mass generated, the balance is lost as heat. The mass of organisms thus generated in turn form the matter and energy sources (prey) for other organisms that live on them (predators), and these in turn become prey to yet other predators. Each prey-predator transformation is accompanied by a substantial loss of energy; in this fashion through the sequential chain of life there is a continuous reduction of the organically bound energy originally fixed by the photo-synthetic autotrophs. When the organic energy reduces to zero heterotrophic life ceases. In wastewater treatment plants the full chain of life does not develop so that a substantial fraction of the input energy is retained in the life mass in the system but this fraction is physically separated from the liquid in the sludge wasted daily (for further treatment) leaving very little organic energy in the effluent.

Presence of organic energy in the receiving body of water gives rise to fungal and other heterotrophic growths, deoxygenation of the water (due to the metabolic activities of these growths) thereby rendering the water unsuitable for higher life forms such as fish, anaerobic conditions causing fermentation and redissolution of heavy metal salts, and so on. These effects reduce the aesthetic appearance, recreational use and the re-use of the water.

(c) Chemical autotrophs derive their energy for growth from the oxidation of inorganic compounds. In wastewater microbiology the principal species of interest in this genera are the nitrifying bacteria. These bacteria are obligate aerobes (i.e. can grow only when oxygen is present), and derive energy by oxidizing saline ammonia to nitrite and nitrate. In wastewater treatment the nitrifiers form a vital link in the removal of the autotrophic element nitrogen from the water by converting ammonia nitrogen to nitrate nitrogen when, by appropriate design, the heterotrophs can be forced to utilize the nitrate in the heterotrophic metabolism of carbonaceous material, in which action the nitrate is reduced to molecular nitrogen and escapes as nitrogen gas. (see later).

1.1 Objectives of wastewater treatment

From the considerations above the objectives of wastewater treatment are seen to be twofold:

1. Reduction of organic-bound energy to a level such that the heterotrophic growth and associated deoxygenation effects in the receiving body of water are acceptably low.
2. Reduction of the autotrophic substances, phosphates, ammonia and nitrates, to levels such that photo-synthetic autotrophic growths and their capacity to fix solar energy as organic energy in the receiving body of water are acceptably low.

2. BIOLOGICAL BEHAVIOUR

Biological metabolism is principally a process of energy conversion. In biological wastewater treatment processes, the input energies are present in the wastewater in the form of carbonaceous and nitrogenous organic compounds. These are converted into other energy forms during which some of the energy is lost as heat.

The carbonaceous energy is utilized by the heterotrophic group of organisms. This is a wide spectrum group, which, given sufficient time and appropriate environmental conditions, will utilize every type of organic material. The group is ubiquitous and in any given situation those members of the group that obtain maximum benefit from the specific organic material and environmental conditions will develop. As the organic source of the environmental conditions change, so associated changes in the heterotrophic organism species take place.

The nitrogenous material serves as prime energy source in the free and saline ammonia form. Proteinaceous nitrogen is broken down into its ammoniacal and carbonaceous constituents by heterotrophs. The free and saline ammonia is utilized as an energy source by two specific organisms species, the *Nitrosomonas* and *Nitrobacter*. The former converts ammonia to nitrite and the latter nitrite to nitrate, these two organism species jointly being called the nitrifiers. Compared to most of the members of the heterotrophic group, the nitrifiers are slow growing. Because the group consists of only two species, each performing a particular function in the nitrification process, the environmental conditions must be conducive to both species, otherwise one or the other will not propagate and nitrification will be adversely affected. In general the environmental conditions for the nitrifiers are much more circumscribed than those for the heterotrophs. With the heterotrophs, the variety of organisms is so large that some group of the species nearly always will adapt to the biodegradable organic material in the influent or changes in the environmental conditions. With the nitrifiers, once the environmental conditions change to beyond a rather restricted range, they cease to grow.

2.1 Electron donors and acceptors

For all the organisms the energy source serves two functions:

(1) for the supply of material which is transformed to new cell material i.e. synthesis of new cell mass;
(2) for the supply of energy to effect the transformation.

For heterotrophic growth energy is obtained from the carbonaceous material, as follows: Each organic molecule is split to give hydrogen ions, carbon dioxide molecules and electrons. Because it releases electrons the organic molecule is called an *electron donor* and on yielding electrons the molecule is said to be *oxidized*. The electrons (and hydrogen ions) are eventually transferred to a molecule that can receive them and this molecule is called an *electron acceptor*; on receipt of elecrons the molecule is said to be *reduced*. In this oxidation-reduction (redox) reaction *free energy* is released, that is, energy that can be utilized to do work.

Under *aerobic* conditions the electron acceptor is oxygen (O_2) and it is reduced to water (H_2O). If no oxygen is available but nitrate (NO_3^-) or nitrite (NO_2^-) is, an *anoxic* state exists; the NO_3^- and NO_2^- serve as electron acceptors and both are reduced to nitrogen gas and water. If no O_2, NO_2^- or NO_3^- is present an *anaerobic* state exists and the organism generates its own electron acceptor internally (endogenously), in contrast to externally provided (exogenous) electron acceptors (O_2, NO_2^- and NO_3^-), provided in the aerobic and anoxic states.*

The quantity of free energy released in a redox reaction depends on the electron donor and the acceptor. When carbonaceous material is the donor and oxygen is the final acceptor the free energy is relatively high; with NO_2^- and NO_3^- as acceptor the free energy is about 5% less than with oxygen (McCarty, 1964). Internally generated electron acceptors give rise to redox reactions that yield very little free energy.

Taking oxygen as the electron acceptor the free energy per electron transferred for different configurations of carbonaceous molecules lies within a relatively narrow band of values. For sewage which contains an innumerable number of different carbonaceous substances, the mean free energy per electron transferred is virtually constant, McCarty (1964).

In aerobic synthesis of organism mass a *fraction* of the carbonaceous material (with oxygen as electron acceptor) is oxidized via a complex set of redox reactions, to yield free energy, see Figure 1.1. The free energy (via ADP-ATP exchanges) is utilized to reorganise the remaining carbonaceous molecules to protoplasmic material, the free energy being lost as heat in this reorganization.

In the qualitative description above the question that arises is: "How does one quantify this synthesis reaction?" The quantification rests on the indestructability (conservation) of matter in chemical reactions.

*In Bacteriology if the terminal hydrogen ion and electron acceptor is molecular dissolved oxygen, then the process of substrate oxidation is termed aerobic; if it is chemically bound oxygen, i.e. nitrate, nitrite or sulphate, the process is termed anaerobic; if the terminal electron acceptors originate outside the cell, i.e. exogenous, the process is termed one of respiration; if the terminal electron acceptors originate inside the cell, i.e. endogenous, the process is termed fermentation. In Sanitary Engineering, the usage of some of these terms differs from that in Bacteriology. In particular if no dissolved oxygen is present but nitrate is, the process of environment is called *anoxic*; if neither dissolved oxygen nor nitrate is present, the process or environment is called anaerobic.

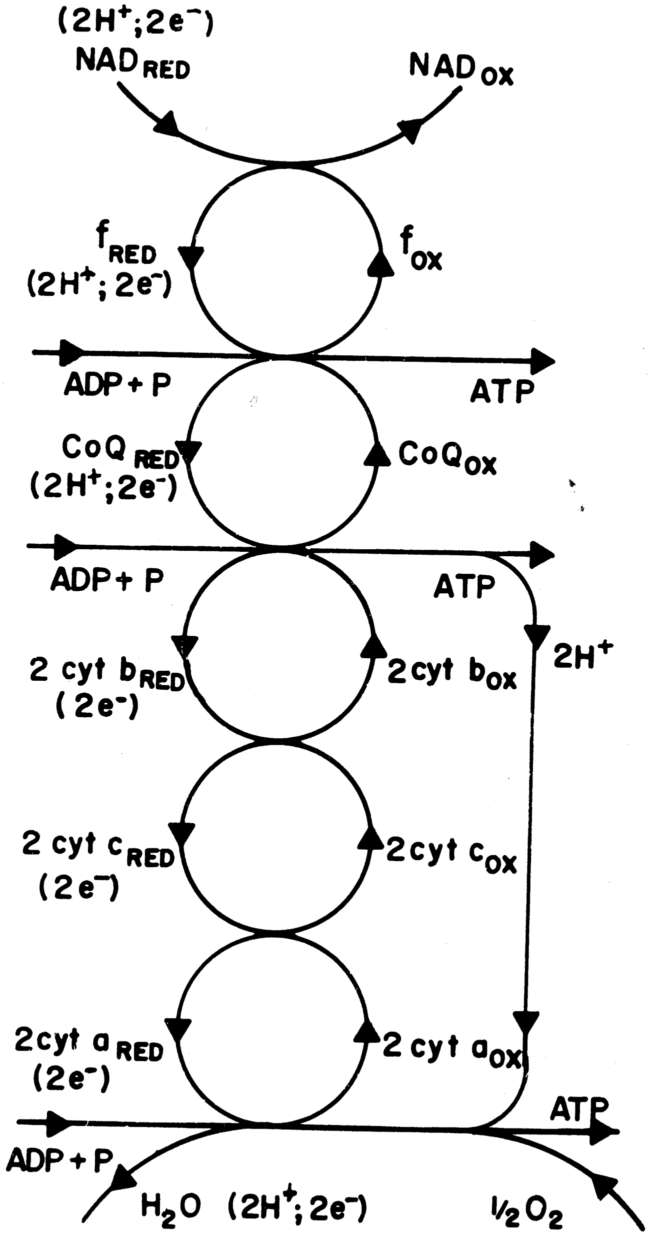

FIGURE 1.1a: ATP (biological energy) formation by sequential oxidation-reduction reactions (e^- transport) in the cytochromes of the micro-organism

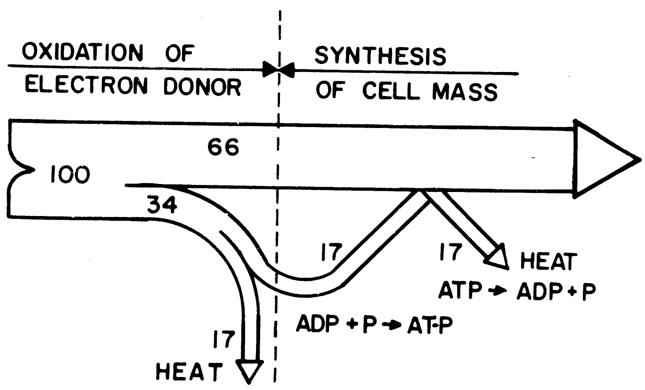

FIGURE 1.1b: Energy change during synthesis

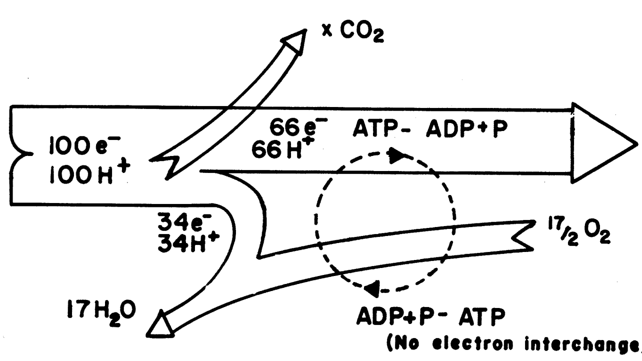

FIGURE 1.1c: Electron changes during synthesis

During synthesis, in the set of redox reactions, electrons and hydrogen ions are transferred from the donor to the acceptor (see Figure 1.1) but, the electrons can be neither created nor destroyed; however, they can be measured, as follows: The number of electrons accepted in the reduction of a molecule of oxygen to water is a defined physical-chemical quantity. If the carbonaceous material is oxidized completely to CO_2, with oxygen as acceptor, the mass of oxygen consumed gives a direct stoichiometric measure of the electrons transferable from the organic molecules, and, as the free energy per electron (with O_2 as acceptor) is virtually constant for all carbonaceous materials, indirectly a measure of the free energy also is obtained. Thus, in a synthesis reaction, by (1) oxidizing completely the input biodegradable material and (2) oxidizing completely the synthesized material, the ratio

$$\frac{\text{oxygen demand of synthesized material}}{\text{oxygen demand of input material}}$$

$$= \frac{O_2 \text{ equivalent of synthesized material}}{O_2 \text{ equivalent of input material}}$$

$$= \frac{\text{electrons available in synthesized material}}{\text{electrons available in input material}}$$

= specific *yield* coefficient of the reaction.

Very closely the specific yield coefficient above also reflects the ratio

$$\frac{\text{free energy in the synthesized material}}{\text{free energy in the input material.}}$$

The difference between (O_2 equivalent of input material) and (O_2 equivalent of synthesized material), reflects the electrons transferred to oxygen in order to release free energy to drive the synthesis reaction i.e.,

$$O_2 \text{ demand} = (O_2 \text{ equivalent of input material}) - (O_2 \text{ equivalent of synthesized material}). \qquad (1.1)$$

This equation is stoichiometrically exact and forms the basis for fundamental investigations into the description of the activated sludge process. Very closely, in fact for all practical purposes exactly, it also reflects the energy changes in aerobic and anoxic redox reactions. Thus, by the developments above, we can trace the energy changes by measuring the electrons (as equivalent oxygen) in the input material and the electrons contained in the synthesized material, the difference between these two being equal to the oxygen utilized during change in state of the material. Electrons are indestructible and can and always should be accounted for in any investigation; this brings to the fore the importance of electron *mass balances* to check the validity of investigations both in research and in the operation of a plant.

During nitrification, ammonia serves as the electron donor and oxygen as the acceptor, to convert NH_4^+ to NO_2^- and NO_3^-. Ammonia is present in the influent either as an NH_3 radical in protein molecules (albuminoid ammonia) or as dissolved ammonia (NH_3) and ammonium ions (NH_4^+) (free and saline ammonia respectively). On conversion of the proteinaceous ammonium radical to the free and saline form (by the heterotrophs) it becomes available as an electron donor with oxygen as acceptor to form NO_3^- and NO_2^-. As the molecular forms of both NH_4^+ and NO_3^- are fixed, once the total mass of nitrogen available for nitrification is known the oxygen necessary to oxidize it is also known, and hence the electron donor capacity.

One last point: The yield as defined earlier, is always an experimentally derived value. Biochemistry is now sufficiently advanced to allow the yield to be calculated theoretically fairly accurately by taking into account the stoichiometry of the electron donor and acceptor, the thermodynamic free energy released per electron transferred and the biochemical pathways through which the transfer takes place (for example, the Embden-Meyerhof pathway – Krebs cycle sequence; Payne, 1971). Such an approach, although of importance conceptionally, is not so valuable practically because in wastewater treatment plants, the microbiological mass does not consist of pure culture but comprises a complex ecological system of primary organisms and ones that prey on them.

3. CARBONACEOUS ENERGY MEASUREMENT

3.1 Chemical Oxygen Demand test

The discussion in the previous section, in which it was shown that the energy changes in biological reactions is reflected in the number of electrons transferred, led to the conclusion that the electron donor capacity can be measured in terms of the oxygen required to oxidize the carbonaceous matter to CO_2. Such a measurement is available in the Chemical Oxygen Demand (COD) test.

In the COD test the electron donor capacity of carbonaceous material is measured by oxidizing the material by a strong oxidant i.e. a hot dichromate sulphuric acid solution, in which the electron acceptor is oxygen. This oxidant is so strong that oxidation of the carbonaceous matter is complete, or very near complete. From the half reaction for the reduction of oxygen,

$$4e^- + 4H^+ + O_2 \rightarrow H_2O$$

it can be seen that 1 mole of molecular oxygen (32 g) is equivalent to 4 electron equivalents. Stoichiometrically in the COD test when the equivalent of 1 g O_2 is used up then the mass of COD oxidized is alslo 1 g, i.e.

$$1 \text{ g COD} \equiv 1 \text{ g } O_2 \equiv 1/8 \text{ electron equivalent.}$$

The COD test, therefore, supplies the information required to trace the electron and associated energy changes in a process.

Ammonia is not oxidized in the test so that the test reflects only the electron donor capacity of carbonaceous material.

In practical application some organic material is not oxidized by the COD test under any circumstances, such as the aromatic hydrocarbons and pyridenes, although these are utilized by micro-organisms. Poor estimates of COD can also arise with lower fatty acids such as acetate because these compounds are in the unionized form at the low pH at which the test is performed – the unionized molecules can be extremely difficult to oxidize; in this regard addition of a catalyst silver sulphate overcomes the problem to a large degree to give virtually 100% oxidation.

In performing a COD test it should be remembered that the test involves a time dependent reaction and the specified refluxing period of 2 hours must be adhered to to ensure complete or near complete oxidation. Furthermore, the degree of oxidation is subject to mass action effects – equivalent masses of the same organic compound refluxed at different final test volumes yield different results for the COD; the temperature at which the refluxing step takes place also affects the oxidation rate, the refluxing temperature in turn is dependent on the concentration of sulphuric acid in the test. These factors have all been considered in the development of the COD test procedure. Hence, the COD test will give reliable results only if the test is done in strict accordance with the set procedures, for example, those in Standard Methods (1981). If these procedures are strictly followed our experience is that the COD test adequately reflects the electron donor capacity of the sample. Application of COD test results to carbonaceous material mass balances on activated sludge plants at steady state also indicate that with careful experimental work mass balances between 98 and 102 per cent are obtainable i.e. the mass of influent COD per day can be accounted for by the sum of the daily mass of oxygen utilized and the daily masses of COD in the waste sludge and effluent. Extensive investigations into COD balances by Schroeter, Dold and Marais (1983) have indicated that where poor balances are obtained, invariably the causes can be traced to errors in oxygen demand for nitrification (to correct the total oxygen demand to the carbonaceous oxygen demand) and/or errors in the oxygen utilization rate measurement.

3.2 Total Oxygen Demand test

In the Total Oxygen Demand (TOD) test the sample is oxidized at high temperature in a combustion oven. The TOD is the mass of oxygen required for the oxidation of all oxidizable material contained in a unit volume of wastewater sample; both carbonaceous and nitrogenous compounds are oxidized. A problem with this test is that the amount of oxygen available in the combustion oven influences the degree of oxidation of ammonia and organic nitrogen – with oxygen in excess there is a conversion of nitrogenous material to nitric oxide, whereas when oxygen is not in excess, oxidation of the nitrogenous material may be partial only. Furthermore, nitrate, nitrite and dissolved oxygen in the sample influence the TOD test result. Because the test determines both the carbonaceous and nitrogenous oxidation potential, an additional TKN test is required to isolate the carbonaceous fraction of the TOD. At the present stage of development of the test, difficulty is found in obtaining representative and reproducible readings for samples which contain particulate matter – these samples must be thoroughly homogenised to obtain representative readings as the sample volume is extremely small ($10-20\ \mu\ell$). Sample homogenisation also increases the dissolved oxygen contamination. The difficulties of applying the TOD test to wastewater can be overcome, but the resulting test procedure reduces its attraction for general routine use.

3.3 Biochemical Oxygen Demand test

This test provides a measure of the oxygen consumed in the biological oxidation of carbonaceous material in a sample over 5 days at 20°C. According to Gaudy (1972) the origins of this test can be traced back about a hundred years to Frankland who appears to have been amongst the first to recognise that the observed oxygen depletion in a stored sample is due to the activity of micro-organisms. The Royal Commission on Sewage Disposal appeared to have developed the concept of using the oxygen depletion as a measure of the strength of pollution. Over the past century extensive research towards elucidating the meaning of the test, quantifying it and integrating it in theoretical descriptions of wastewater treatment processes, has been undertaken. To date the Biological Oxygen Demand (BOD) test is still the most popular parameter for assessing the pollution strength of influents and effluents and for describing the behaviour of treatment processes. Its continued popularity seems to stem mainly from the body of experience that has built up in its use. Whereas there is justification for retaining it in effluent descriptions for regulatory purposes, in so far as present day scientific descriptions of treatment process is concerned few cogent arguments for its use over that of the COD test can be produced.

The main deficiency of the BOD test arises from the inadequacy with which it assesses the electron donor capacity of the carbonaceous organic material in a sample. At the end of 5 days some of the biodegradable carbonaceous material originally present in the sample remains as unbiodegradable material. Consequently, it is not possible to perform a mass balance even with respect to the biodegradable input material. Furthermore unbiodegradable material originally present in the sample remains unaffected by the test so that there is no procedure, related to the test, whereby an assessment of this fraction can be obtained. (In contrast in the COD test, because the total electron donor capacity is measured, even though the unbiodegradable fractions are not isolated in the test, it is possible to estimate these fractions from the test results and the kinetic behavioural patterns of the treatment process).

Despite the deficiencies listed above, if the BOD_5 value constituted a *consistent* estimate it could still have provided a relative measure of practical usefulness, but even this cannot be guaranteed. Gaudy (1972) in a thought provoking review on the BOD gave a summary of the work on the BOD test (in which he took a leading part). He concluded that the standard approach to formulating the BOD time curve as a first order reaction is quite invalid: The actual curve arises from two main effects, (1) bacterial synthesis from the biodegradable input – a reaction usually complete in one to two days, and (2) predator growth using the synthesized bacteria as substrate. The second phase, being dependent on the first, usually lags the first so that a distinct "pause" or "plateau" in the time curve is exhibited (see Figure 1.2). The occurrence and duration of this pause is dependent on the bacteria-prey relationship initially pre-

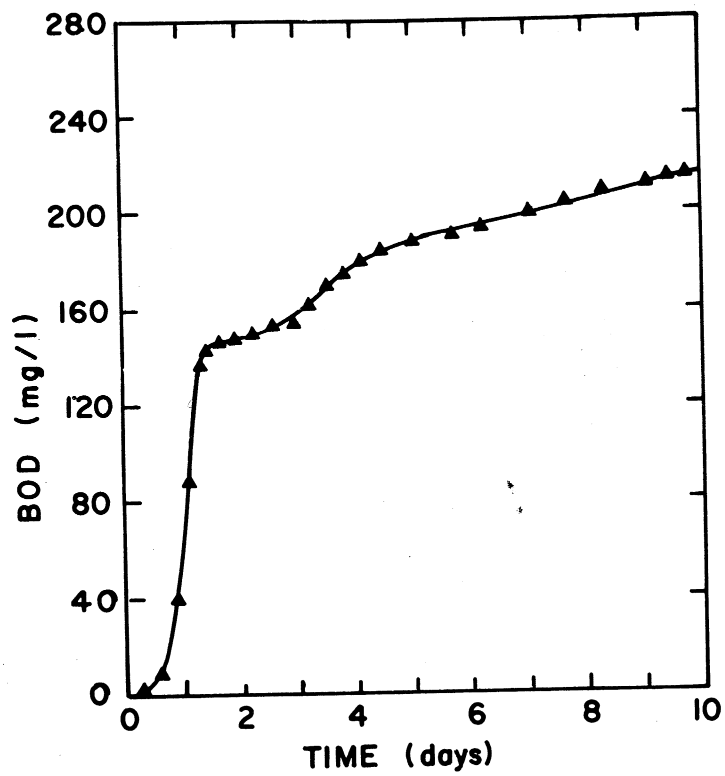

FIGURE 1.2: Typical experimental BOD time curve on wastewater sample with heterogeneous seed clearly showing the "plateau" behaviour after about 1,5 days

sent in the sample, the "strength" of the organic material, presence of inhibitory substances and so on. These compound effects certainly are not reflected by a first order formulation. Furthermore, should nitrification occur during the 5 days the BOD value can be completely misleading; this may not be apparent unless procedures to suppress nitrification are imposed on the test. Evidently the BOD_5 value is subject to a number of influences which can be present in varying degrees resulting in unknown variable effects on the observed value.

Perhaps, the neatest summing up of the use of the BOD test is that due to Hoover, Jasewicz and Porges (1953): "The BOD test is paradoxical. It is the basis of all regulatory actions and is used routinely in almost all control and research studies on sewage and industrial waste treatment. It has been the subject of a tremendous amount of research, yet no one appears to consider it adequately understood or well adapted to his own work". Considered in the light of the biochemical descriptions of biological growth these remarks are still pertinent today.

3.4 Total Organic Carbon test

In the Total Organic Carbon (TOC) test the sample is oxidized in a combustion chamber in the presence of oxygen and the carbon dioxide content of the burnt gas is measured. If the organic carbon content only is required, then it is necessary to have a pre-treatment stage in sample preparation by acidification and CO_2 purging by sparging, or, to do an additional test in which the inorganic carbon only is measured. The older instruments can only measure the carbon content of soluble organic material but newer models can deal with particulate organic material.

As a procedure to estimate the electron donor capacity of the sample the TOC test can be very misleading. This arises from the fact that the ratio of organic carbon to electrons is not constant for different organic materials. For example taking glucose and glycerol:

The number of available electrons may be determined by

(i) measuring the number of moles of O_2 consumed in the complete combustion of one mole of the compound of interest, and

(ii) multiplying by four (which is the relative number of electrons required to reduce one mole of oxygen).

e.g. (a) Glucose, $C_6 H_{12} O_6$

$C_6 H_{12} O_6 + 6\, O_2 \rightarrow 6\, CO_2 + 6\, H_2O$

6 moles $O_2 \Rightarrow 6 \times 4 = 24\, e^-$ compound

i.e. $24/6 = 4\, e^-$ available per unit organic C.

(b) Glycerol, $C_3 H_8 O_3$

$C_3 H_8 O_3 + 7/2\, O_2 \rightarrow 3\, CO_2 + 4\, H_2O$

7/2 moles $O_2 \Rightarrow \frac{7}{2} \times 4 = 14\, e^-$ compound

i.e. $14/3 = 4{,}66\, e^-$ available per unit organic C.

Comparing glucose and glycerol the electron donor capacity per unit C differs by 17%.

The analysis above points to the inadvisability of utilizing the TOC test to describe the energy based behaviour of wastewater treatment processes. This does not mean that the test does not have other uses; it serves a valuable function as control parameter or as a parameter in tertiary treatment where the carbon content in the effluent may be of great importance.

4. COD/VSS RATIO

Consider a completely biodegradable soluble substrate, for example glucose. If a sample of glucose solution is innoculated with bacteria, syntheses of new bacterial mass will occur. The change in soluble COD, ΔCOD (soluble), will be reflected in an increase in the COD of the bacterial mass, ΔCOD(bacteria), and oxygen utilized to generate free energy, ΔO_2(utilized),

$$\Delta COD(soluble) = \Delta COD(bacteria) + \Delta O_2(utilized) \quad (1.2)$$

Equation (1.2) reflects the destination of the electrons from the donor to the synthesized material and the electron acceptor, oxygen. From extensive work reported in bacteriology, reviewed by Payne (1971), the fraction of ΔCOD(soluble) appearing as ΔCOD(bacteria) appears to be nearly constant. Usually the fraction is expressed as a ratio called the specific yield coefficient Y_{COD} (see Eq 1.1) i.e.

$$\Delta COD(bacteria)/\Delta COD(soluble) = Y_{COD} \quad (1.3)$$

Equation 1.2 can be rewritten in terms of Y_{COD}

$$\Delta COD(soluble) = Y_{COD} . \Delta COD(soluble) + \Delta O_2 \quad (1.4)$$

and recasting Eq (1.4)

$$\Delta O_2 = (1 - Y_{COD})\, \Delta COD(soluble) \quad (1.5)$$

In wastewater treatment kinetics the usage has arisen, probably from the work of bacteriologists, to measure the mass of *volatile solids generated* in the synthesis reaction, X_a, rather than the COD of the volatile solids generated, ΔCOD(bacteria). To retain the usefulness of Eq (1.4), in particular in the form given by Eq (1.5), it is necessary to relate ΔX_a to ΔCOD(bacteria). The usage has developed to express the relationship between ΔX_a and ΔCOD(bacteria) by the ratio of the two parameters, called the COD/VSS ratio, defined by the symbol f_{cv},

$$f_{cv} = COD/VSS = \Delta COD(bacteria)/\Delta X_a$$

Using this definition the yield coefficient Y_{COD} is expressed in terms of a specific yield coefficient with respect to volatile solids Y_h, and the COD/VSS ratio, f_{cv},

$$Y_{COD} = f_{cv} . Y_h \quad (1.6)$$

and inserting in Eq (1.5),

$$\Delta O_2 = (1 - f_{cv} Y_h)\, \Delta COD(soluble) \quad (1.7)$$

Equation (1.7) is very important in biological wastewater treatment kinetics.

Considerable research has been undertaken to establish a value for f_{cv} and to determine the factors affecting it. Theoretically f_{cv} has been evaluated by accepting an empirical stoichiometric formulation for biological sludge, for example that of Hoover and Porges (1952),

$$C_5H_7O_2N + 5O_2 \rightarrow 5CO_2 + 2H_2O + NH_3$$

Stoichiometrically,

$$113 \text{ g VSS} \rightarrow 160 \text{ g O} \rightarrow 160 \text{ g COD}$$
$$\text{i.e. } 1 \text{ g VSS} \rightarrow 1{,}42 \text{ mg COD}$$
$$\text{i.e. COD/VSS} = 1{,}42 \text{ mg COD/mg VSS}$$

Eckenfelder and Weston (1956) presented experimental data that supports this theoretical ratio. However, other theoretical ratios have been proposed and each find some experimental confirmation, see Table 1.1. The basic difficulty in evaluation of f_{cv} arises from the fact that f_{cv} is a ratio so that the small random errors in either the COD or the VSS magnify the dispersion of the ratio; a reasonably stable mean estimate of the f_{cv} value of a sludge requires at least thirty COD and VSS pair determinations. With smaller numbers the mean f_{cv} value can still vary significantly.

TABLE 1.1 THEORETICAL COD VALUES FOR VARIOUS EMPIRICAL FORMULATIONS FOR MICROBIAL SLUDGE (AFTER McCARTY, 1964).

Bacterial formulation	Molecular weight	Theoretical COD in grams			
		Per mole	Per g TSS	Per g VSS	Per g carbon
$C_5H_7O_2N$	113	160	1,28	1,42	2,67
$C_5H_9O_3N$	131	160	1,10	1,22	2,67
$C_7H_{10}O_3N$	156	232	1,33	1,48	2,76
$C_5H_8O_2N$	114	168	1,32	1,47	2,80

Considering the COD/VSS ratio of sludges derived from activated sludge processes treating municipal effluents, it should be remembered that not only is active volatile mass generated but volatile mass is also present in unbiodegradable particulate form in the influent (which accumulates in the sludge mass) and inert biological residue generated due to endogenous respiration (which also accumulates in the sludge mass). Consequently the sludge normally consists of three fractions, active, inert influent and endogenous. Each of these could have a different f_{cv} value, a factor not considered in deriving the theoretical ratios discussed earlier. The proportions of the three fractions in a sludge depend on the sludge age, the active fraction being relatively small at long sludge ages and high at short sludge ages. For practical purposes an important aspect, therefore, is whether f_{cv} changes with sludge age or whether a mean experimentally derived value can be assigned to all activated sludges. Accordingly Marais and Ekama (1976) and Schroeter, Dold and Marais (1982) investigated the COD/VSS ratio of activated sludge on laboratory scale units treating municipal wastewater for sludge ages ranging from 2 to 30 days.

Their data indicate that despite the significant changes in proportion of the three fractions with changes in sludge age, f_{cv} appeared to remain substantially constant, at 1,48 mgCOD/mgVSS. This finding is very important because it allows that COD balances generally can be performed using VSS measurements on the waste sludge.

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